Solar Energy Assisted Purification of Water using Nanocrystalline Stratified Transition Metal Oxide Semiconductor Thin Films"

by

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Radiant energy from the sun is vital for life on our planet. It determines the surface temperature of the earth as well as supplying virtually all the energy for natural processes both on the earth and in the atmosphere. One of the most important problems affecting people throughout the world is inadequate access to potable water. Over one billion people are exposed to unsafe drinking water due to poor source water quality and lack of adequate water treatment, a problem expected to grow worse in the coming decades, with water scarcity occurring globally, even in regions currently considered water rich. The problem is particularly significant in developing countries and in arid areas where water sources are scarce. This is a continually growing problem due to rising population and increasing demands for source of waters. Moreover, with rapid industrialization, many pollutants such as various dyes, sulphates and toxic compounds are dumped into the water. Waste water from textile processing plants, is highly colored and difficult to decolorize. Azo- dyes form the argest group among the synthetic colorants (60–70%). Their chromophoric system consists of azo groups (–N N–) in association with aromatic systems and auxochromes (–OH, –NH2, etc.). It was estimated that more than 50,000 tons of dyes are discharged from dyeing and coloration industries every year. Since azo dyes are intentionally designed to resist degradation, it is very difficult to degrade them by conventional wastewater treatment methods. In addition, the toxicity, mutagen city and carcinogenicity of azo-dye degradation byproducts are of great environmental concern. Chemical disinfection options such as chlorine and iodine treatment, ozonization, flocculation and filtering require additives and are also not cost-effective. Physical treatment such as boiling, distillation, reverse osmosis and UV irradiation require considerable energy input. One alternative for drinking water treatment method that has been proposed is solar photocatalytic purification, a process that could be utilized in countries that receive abundant sunlight. Wide band gap semiconducting transition metal oxide photocatalysts have been shown to decompose organic compounds, bacteria and other impurities present in water. The most prominent, widely studied photocatalyst is WO_3/TiO_2. Upon irradiation of a semiconductor with photons of energy higher than the band gap, charge carriers are generated. Those which are not lost by recombination can carry out electron transfer reactions at the interface between the semiconductor and a liquid containing an oxidizable or a reducible species. In the case of an n-type oxide semiconductor, valence band holes (h+) lead either directly, or via OH^- radicals, to the oxidation of the solute (D), while conduction band electrons (e^-) lead to reduction of dissolved oxygen, if present, forming superoxide radical anions, $^-O2^-$ and OH radicals, if not participating in the degradation process, finally lead to the formation of water and oxygen, respectively. Recombination of photogenerated positive holes and electrons inside the semiconductor are responsible for a rather low quantum yield of the photocatalytic degradation. A way of increasing electron– hole separation and consequently enhancing quantum yield is the use of transition stratified metal oxide semiconductor thin films and an

application of electrical bias, which is possible when the photocatalyst is deposited on an electrically conducting substrate. Here, the working point of the photoelectrochemical reaction can be shifted to the potential range where photon flux limited (plateau) currents flow. Due to backside illumination through the transparent conducting substrate, direct photolysis is avoided and even strongly colored solutions can be treated, as actinic light is encountering the semiconducting layer first. Electrons are drawn away from the interface, and reduction reaction occurs at the counter electrode. Oxygen is available in sufficient quantity as a depolarizer for the counter electrode, firstly due to rapid recirculation of the solution through a reservoir in contact with air and, secondly, due to water oxidation by reaction steps initiated by valence band holes. As to the organic molecules which can be oxidized in this way, azo-dyes are of special interest, as mentioned above. Different methods have been explored for the removal of azo-dyes from solution. Bourikas et al. investigated the adsorption of a textile azodye, acid orange 7 (AO7), on the surface of WO_3/TiO_2 in aqueous suspensions in the range of 0.1–1 mM at pH values between 2 and 10. Liu et al. studied degradation of AO7 quite differently, using a Fe/granular activated carbon system in the presence of ultrasound. Fernandes et al. studied the electrochemical degradation of AO7 on a boron doped diamond electrode (BDD) and obtained 98% and 77% color and chemical oxygen demand (COD) removal, respectively. AO7 photofading by a photocatalytic method using WO_3 /TiO_2 nanotubes was also reported. Here photoelectrochemical degradation of organic compounds by TiO2, taking AO7 as a model compound. The main aspects studied are (i) the use of electrical bias, (ii) the use of large catalyst surfaces, and (iii) the possibility of using solar light that be used efficiently concentration of the contaminant down to small concentration.